

AI-oriented smart energy management systems and load forecasting research

AI for Smart Energy Management and Load Forecasting: Key Insights

AI is now central to smart grids and microgrids, enabling accurate load and renewable generation forecasting, real-time control, and intelligent demand response. Research spans deep learning architectures, IoT-enabled frameworks, and energy management systems from buildings up to regional grids.

Core Forecasting Methods and Performance

- **Deep learning dominance:** LSTM, GRU, CNN, and hybrid LSTM–CNN models usually outperform classical ML (RF, SVR, XGBoost) and statistical models (ARIMA) for short- and medium-term load forecasting (Shirzadi et al. 2021, 27; Zhang et al. 2023; Ibrahim et al. 2022; Hafeez et al. 2020, 114915; Cordeiro-Costas et al. 2023; Abumohsen et al. 2023; Tarmanini et al. 2023; Farsi et al. 2021, 31191-31212; Bouktif et al. 2018; Bedi and Toshniwal 2019; Butt et al. 2020, 400-425 ; Islam and Ahmed 2022).
- Reported accuracies: $R^2 \approx 0.90$ – 0.96 and MAPE as low as 2.9–4% in STLF and regional forecasts, clearly better than AdaBoost, ARIMA and other baselines (Shirzadi et al. 2021, 27; Ibrahim et al. 2022; Abumohsen et al. 2023; Tarmanini et al. 2023; Bouktif et al. 2018; Bedi and Toshniwal 2019).
- Deep networks are especially effective for **nonlinear, seasonal, and long-dependency** patterns (Aslam et al. 2021; Shirzadi et al. 2021, 27; Hafeez et al. 2020, 114915; Farsi et al. 2021, 31191-31212; Bouktif et al. 2018; Bedi and Toshniwal 2019; Butt et al. 2020, 400-425).

Main AI Forecasting Approaches

Approach	Typical use	Notes	Citations
LSTM / RNN / GRU	STLF, MTLF, building & regional loads	High accuracy, handles sequences	(Shirzadi et al. 2021, 27; Ibrahim et al. 2022; Cordeiro-Costas et al. 2023; Abumohsen et al. 2023; Farsi et al. 2021, 31191-31212; Bouktif et al. 2018; Bedi and Toshniwal 2019; Butt et al. 2020,

400–425

)| | CNN or Conv-1D | Feature extraction + sequence models | Strong in hybrids (CNN–LSTM, PLCNet) | (Zhang et al. 2023; Cordeiro-Costas et al. 2023; Farsi et al. 2021, 31191-31212; Han et al. 2021, 3170-3179; Butt et al. 2020, 400-425)| | Hybrid DL + optimization | LSTM/other DL + GA, heuristic tuning | Improves lags, architecture, hyperparameters | (Hafeez et al. 2020, 114915; Han et al. 2021, 3170-3179; Bouktif et al. 2018; Butt et al. 2020, 400-425)| | Classical ML (RF, SVR, XGBoost)| Baselines, sometimes competitive | Generally lower accuracy than DL | (Shirzadi et al. 2021, 27; Ibrahim et al. 2022; Cordeiro-Costas et al. 2023; Tarmanini et al. 2023; Bouktif et al. 2018; Forootan et al. 2022)|

FIGURE 1 Comparison of common AI forecasting model families

AI-Oriented Smart Energy Management Architectures

- **IoT + DL frameworks:** Edge/cloud systems collect smart-meter and sensor data, run DL models for short-term forecasts, and coordinate demand–response and dispatch (Singh et al. 2025; Han et al. 2021, 3170-3179; T et al. 2024).
- ORA-DL integrates deep neural networks, reinforcement learning, and multi-agent control for **real-time resource allocation**, improving prediction accuracy (~93%), grid stability (~96%), and cutting costs by ~23% (Singh et al. 2025).
- Edge-intelligence frameworks place forecasting models on resource-constrained IoT devices, supervised by cloud servers for anomaly detection and grid coordination (Han et al. 2021, 3170-3179).

Demand Response, HEMS, and Advanced Control

- AI supports **demand response** by selecting users, learning consumption behavior, optimizing prices, and controlling devices in near real time (Khan et al. 2023, 1477-1505; Shareef et al. 2018, 24498-24509).
- Home Energy Management Systems use AI (ANN, fuzzy logic, ANFIS, heuristic optimizers) to schedule appliances under DR programs, balancing cost, comfort, and emissions (Shareef et al. 2018, 24498-24509).
- Reinforcement learning is used for **microgrid and PV management**, dynamically optimizing local energy use and storage (Singh et al. 2025; T et al. 2024; Khan et al. 2023, 1477-1505).

Challenges and Research Gaps

- High data requirements, need for high-performance/edge computing, and integration of heterogeneous IoT data remain key issues (Aslam et al. 2021; Singh et al. 2025; Hafeez et al. 2020, 114915; Han et al. 2021, 3170-3179; Forootan et al. 2022).
- Uncertainty from renewables, weather, and outages still complicates forecasting and real-time control, motivating probabilistic and robust AI methods (Aslam et al. 2021; Percuku et al. 2025; Forootan et al. 2022; Khan et al. 2023, 1477-1505).

Summary

AI-based forecasting, especially deep learning, consistently improves accuracy for short- and medium-term load and renewable generation prediction, enabling better planning, scheduling, and demand response. When embedded in IoT-centric, edge/cloud architectures and combined with reinforcement learning and optimization, these models underpin intelligent energy management from homes to smart cities. Remaining work focuses on data efficiency, real-time deployment, and handling uncertainty at scale.

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